

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I (Introduction to Optical Fibers)

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: Design of Optical fibers

Name of the Innovative Practice: COMSOL Multiphysics Simulation

Date & Duration: 17.07.2019& 50 Minutes

ICT Tool Used: Computer, COMSOL Multiphysics software

Description:

COMSOL Multiphysics is cross-platform finite element analysis, solver and multiphysics simulation software. It allows conventional physics-based user interfaces and coupled systems of partial differential equations (PDEs). COMSOL provides an IDE and unified workflow for electrical, mechanical, fluid, acoustics and chemical applications. Users may use drag-and-drop tools (Form Editor) or programming (Method Editor). COMSOL Server is a distinct software for the management of COMSOL simulation applications in companies. Several modules are available for COMSOL categorized according to the applications areas Electrical, Mechanical, Fluid, Acoustic, Chemical, Multipurpose, and Interfacing

Goals (Learning Outcomes):

The students will be able to familiarize the students about the COMSOL Multiphysics software and to design an optical fiber.

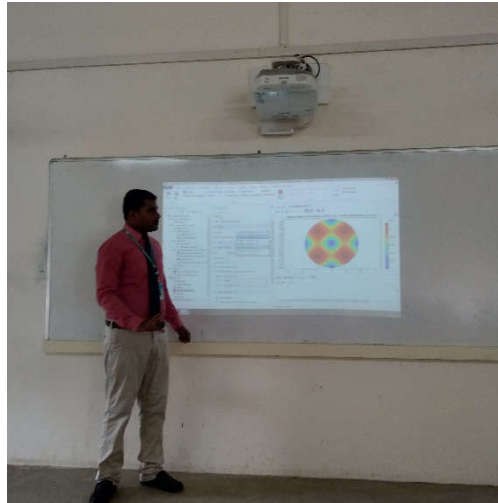
Use of appropriate methods:

Justification for choosing the topic using simulation activity:

Students can able to design optical fibers with various design parameters and also they can calculate the attenuation losses, bending losses and visualisation of fundamental mode.

Effective presentation

Details of the Implementation:



Reflective Critique:

- **Benefits:**

The students were interested in learning the new software COMSOL Multiphysics. They can visually view the fundamental modes and calculation of various losses associated with optical fibers.

- **Challenges:**

Initially students were confusing the procedures and configuration to design a step index fiber. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same through demonstration, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	2	-	-	1	-	1	-	-

CO1: After completion of the course, it is expected that the students will be able to apply the fundamental concept of optical fiber modes and their configurations in telecommunication systems.

References:

- Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
- <https://www.comsol.com/blogs/step-index-fiber-simulation>

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UNIT II (Signal Degradation Optical Fibers)

Degree, Semester & Branch: VII Semester B.E ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: Bending Loss and Connector Loss

Name of the Innovative Practice: Hardware Demonstration

Date & Duration: 23.07.2019 & 30 Minutes

ICT Tool Used: Optical Trainer Kit

Description:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

- The students will be able to know about the concepts of various losses that occur in the optical fiber communication system.

Use of appropriate methods:

Justification for choosing the topic using demonstration activity:

Demonstration of the experiment shows clear idea about the concepts. The students will be able to gather more information by doing the experiment in the kit rather than explaining the concepts theoretically.

Effective presentation

Details of the Implementation:



Reflective Critique:

- **Benefits:**

The students were interested in learning the concepts if we visually show the experiments. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of bending loss and connector loss more clearly by performing demonstration.

- **Challenges:**

Basically some time is needed to explain the basic concepts of bending loss and connector loss. It was difficult to explain to the slow learners initially, later on after explaining the same through demonstration, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2					1			1		1

CO2: The students will be able to analyze the various signal degradation factors associated with optical fiber.

Reference:

- Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
- <http://www.benchmark-electronics.com/optical-fibre-trainer.html>

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UNIT III (FIBER OPTICAL SOURCES AND COUPLING)

Degree, Semester & Branch: VII Semester B.E ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: Laser Fundamentals

Name of the Innovative Practice: Lecture Video

Date & Duration: 16.08.2019& 50 Minutes

ICT Tool Used: Computers and projector

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

The students were able to understand the concepts of Laser characteristics clearly by the lecture video.

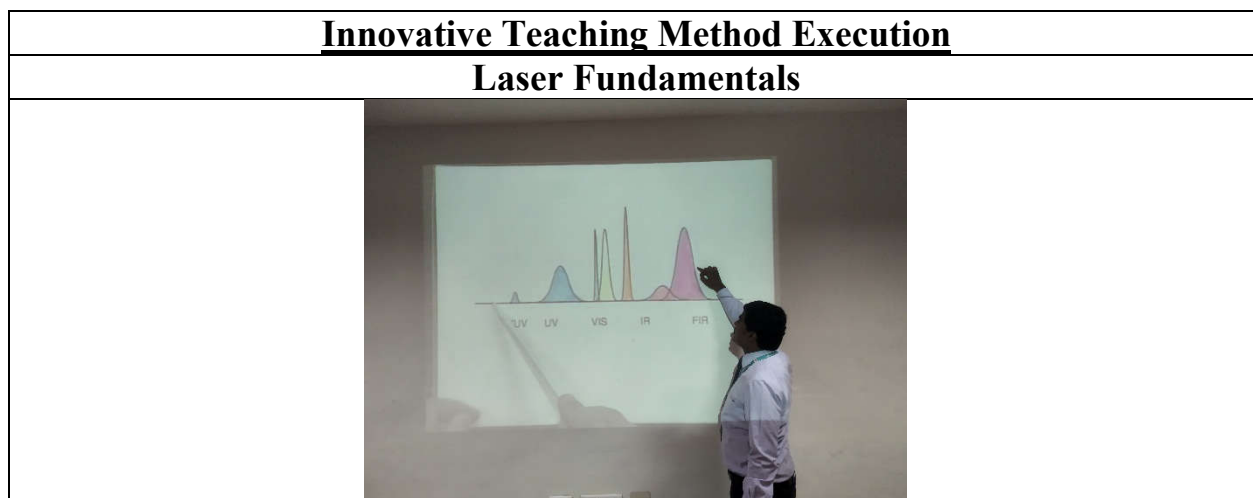
Use of appropriate methods:

Justification for choosing the topic using demonstration activity:

Lecture video shows clear idea about the concepts. The students will be able to gather more information by visualizing the concepts in the screen rather than explaining the concepts theoretically.

Effective presentation

Details of the Implementation:



Reflective Critique:

- **Benefits:**

The students were more interested in learning the concepts if we visually show the videos. They learnt the concepts of laser fundamentals more clearly by seeing the videos. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Some extra time is required in the class room to show the video.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	1							1	1	

CO3: The students will be able to explain the various fiber optical sources and coupling schemes and their use in the optical fiber communication system

Reference:

- Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
- <https://nptel.ac.in/courses/104104085/>

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UNIT IV (FIBER OPTIC RECEIVER AND MEASUREMENTS)

Degree, Semester & Branch: VII Semester B.E ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: Fiber Optic Receiver

Name of the Innovative Practice: One Minute Paper

Date & Duration: 03.09.2019 & 1 Minute

ICT Tool Used: Computer and Projector

Description:

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Students must first organize their thinking to rank the major points and then decide upon a significant question. Sometimes, instead of asking for the main point, a professor may wish to probe for the most disturbing or most surprising item. It is thus a very adaptable tool.

Goals (Learning Outcomes):

The students were able to understand the concepts of various fiber optic receivers and also about preamplifiers like low impedance, high impedance and trans impedance amplifiers.

Use of appropriate methods:

Justification for choosing the topic using one minute paper activity:

The students were more interested in writing the answers for the questions that have been asked since it is a one minute activity. The students could think and recollect the points that have been discussed in the class room.

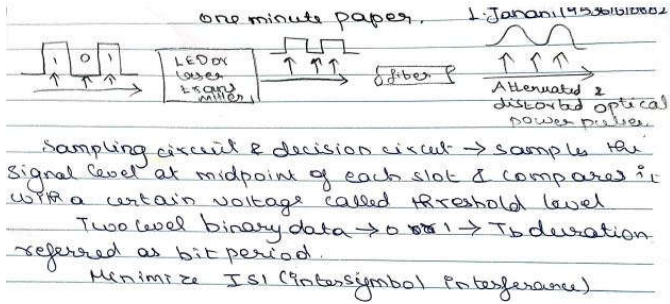
Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution

Fiber Optic Receiver

one minute paper, 1-Janani 195361610024



Sampling circuit & decision circuit → sample the signal level at midpoint of each slot & compare it with a certain voltage called threshold level.

Two level binary data → 0 & 1 → To deviation referred as bit period.

Minimize ISI (Inter-Symbol Interference)

V-Harsheni (95361610024)

Optical Communication and Network

Expt. No.1 one minute paper Page No.1

Fundamental Receiver operation:

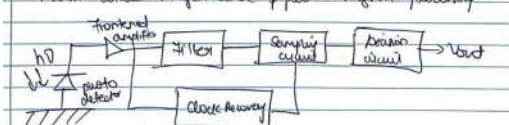
Electric pulse → LED or laser → optical power pulse → attenuated optical

↓

PN → Electrical current → Amplifier → voltage pulse

↓

Decision circuit → Regenerated op pulse → signal processing



Reflective Critique:

- **Benefits:**

The students were more interested in writing the answers for the objective type questions rather than writing theories. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Some of the students felt some time can be given at the beginning of this activity so that they can recollect the points that have been discussed in the class room.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2							1		

CO4: The students will be able to apply various fiber measurement techniques to measure the optical fiber systems performance

Reference:

- Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
- <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>

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UNIT V (OPTICAL NETWORKS AND SYSTEM TRANSMISSION)

Degree, Semester & Branch: VII Semester B.E ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: SONET

Name of the Innovative Practice: Flipped Classroom

Date & Duration: 24.09.2019 & 50 Minutes

ICT Tool Used: Computer and Projector

Description:

A flipped classroom is an instructional strategy and a type of blended learning that reverses the traditional learning environment by delivering instructional content, often online, outside of the classroom. It moves activities, including those that may have traditionally been considered homework, into the classroom.

Goals (Learning Outcomes):

The students will be able to understand the concepts of SONET/SDH and its frame structure and its usage in practical applications.

Use of appropriate methods:

Justification for choosing the topic using flipped class room activity:

The students were given assignment on the particular topic some few days back and they were asked to discuss the same topic inside the class room. They could be able to deliver more concepts. Since the assignment was given some days back, they got time to prepare and discuss efficiently in the class room.

Effective presentation

Details of the Implementation:

<u>Innovative Teaching Method Execution</u>	
SONET	
	

Reflective Critique:

- **Benefits:**

The students were more interested in learning the concepts if they discuss themselves. They learnt the concepts of SONET more clearly by discussing the concepts.

- **Challenges:**

Basically some time is needed to explain the basic concepts of flipped class room. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	1	2	2		1					1		1

CO5: The students will be able to analyze the various optical fiber networks such as SONET/SDH, WDM and Optical CDMA and to measure the network performance

Reference:

- Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
- <https://facultyinnovate.utexas.edu/flipped-classroom>
- https://www.youtube.com/watch?v=qdKzSq_t8k8